IE301 - Election Logistics

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PROBLEM STATEMENT AND VISUALIZATION

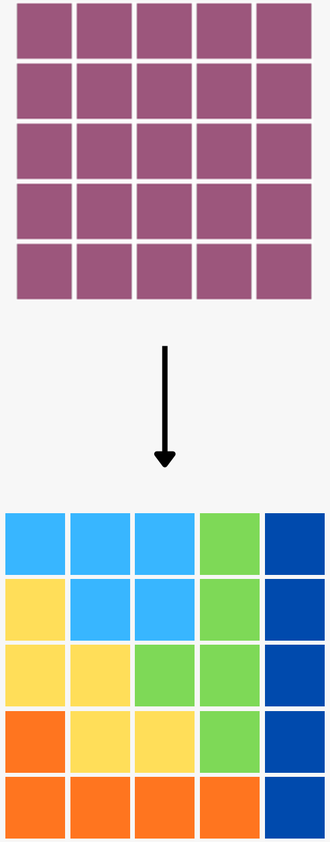
- How should a geographic region be divided into electoral constituencies while ensuring equitable representation and practical accessibility?

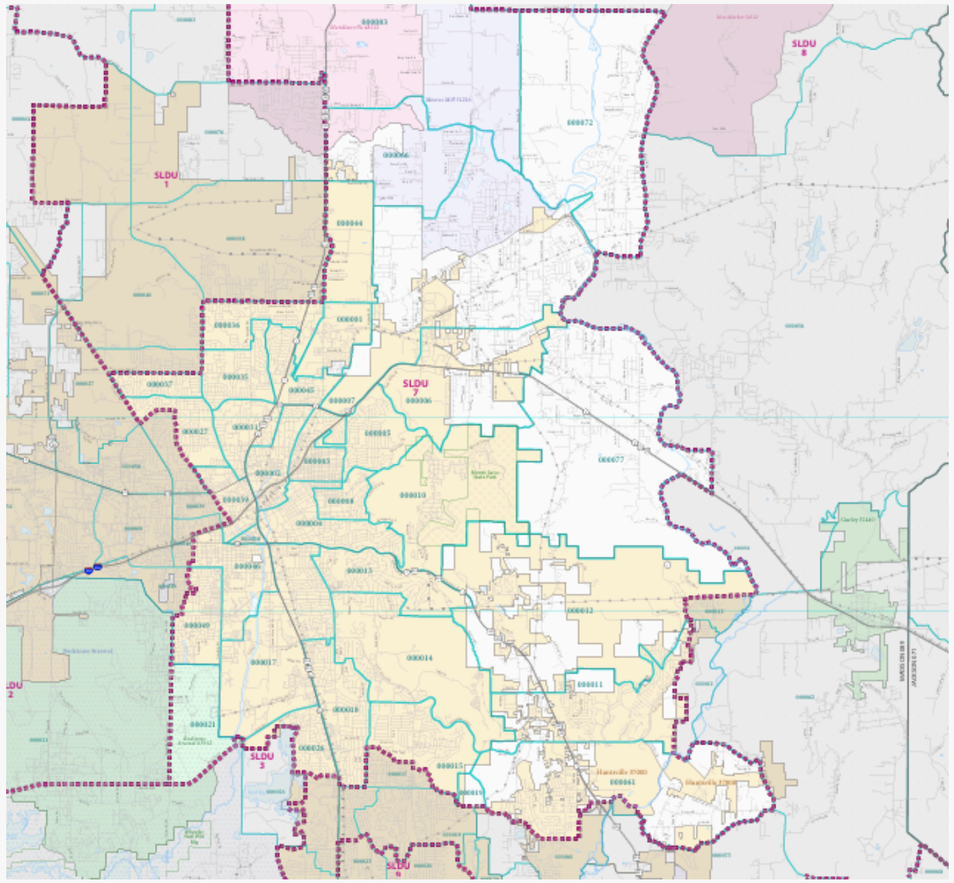
PRIMARY: CONSTITUENCY DESIGN

POPULATION
GEOGRAPHY
DEMOGRAPHICS

SECONDARY: OPERATIONAL

CONSTRAINTS
BOOTH CAPACITY
BUDGET
TRAVEL TIME





Visual references: MGGG, Redistricting a 5×5 Grid; U.S. Census Bureau, 2020 Census Redistricting Data Map Suite

INDIA'S DELIMITATION PROCESS: 1952–PRESENT

Districting means partitioning base geographic units into a fixed number of balanced, contiguous zones, each served by one representative/agent.

1952
- 494 SEATS
- BUILT FROM BASE - SC/ST RESERVATION INTRODUCED

→

1963
- 522 SEATS
- WORKED ON LINGUISTIC STATE REORGANIZATION

→

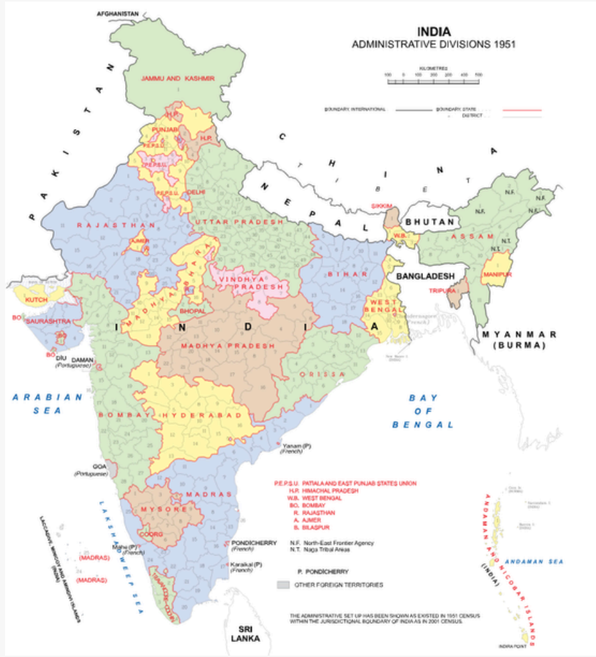
1973
- 543 SEATS
- LAST SEAT INCREASE — STILL THE COUNT TODAY

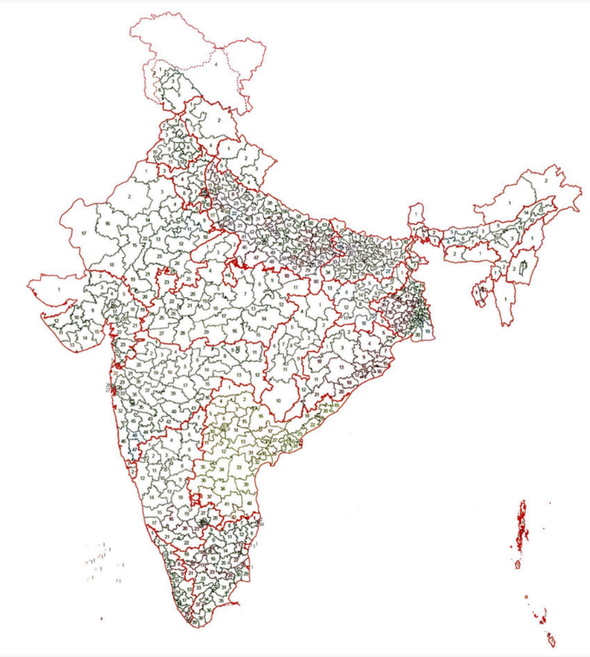
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2002
- 543 SEATS (UNCHANGED)
- ONLY INTERNAL BOUNDARIES WITHIN STATE REDRAWN

FACTORS CONSIDERED

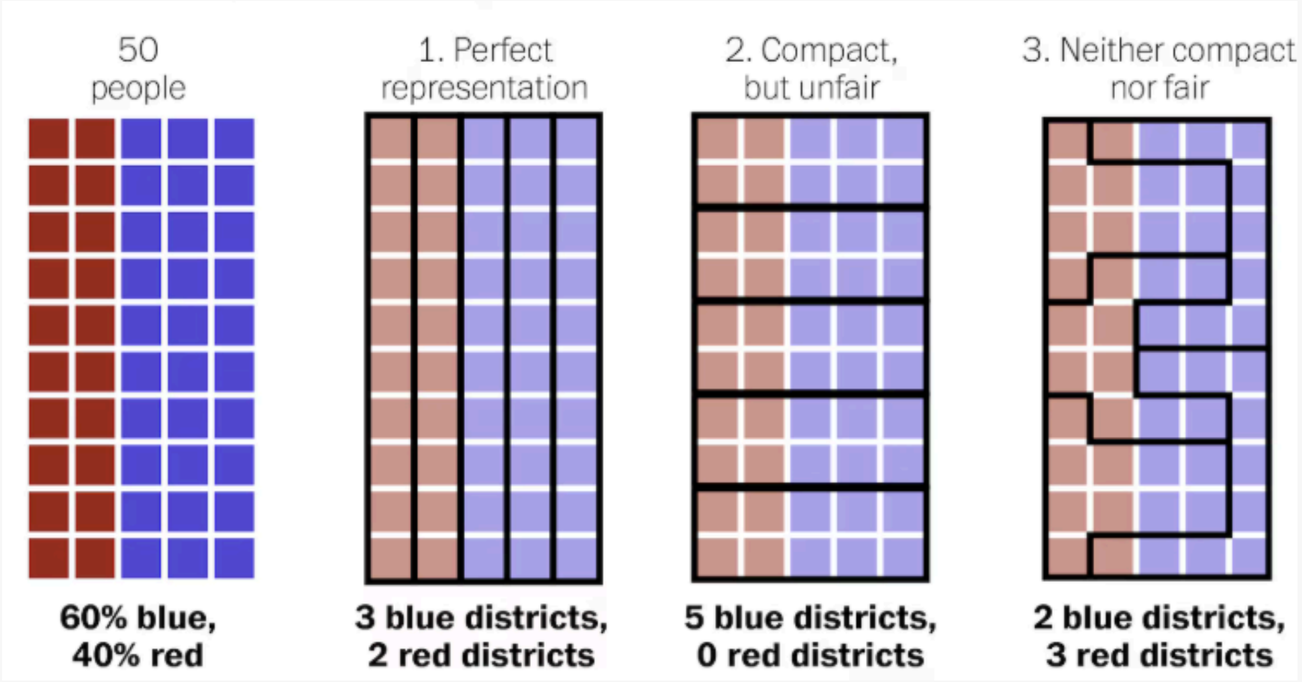
- **Population equality** - each representative should serve roughly equal numbers of people
- **Compact areas and Contiguity** - Administrative practicality, voter accessibility, regular shapes of constituencies
- **Administrative boundary** - Keeps electoral boundaries aligned with existing revenue/governance units
- **Public convenience** - Travel distance, natural features (rivers,terrain,etc.)
- **SC/ST proportional reservation** - Guarantee minimum representation





THE GERRYMANDERING PROBLEM

How can the same population be divided into connected, approximately equal-population districts that produce very different electoral outcomes?



- **WHAT IS GERRYMANDERING ?**
  GERRYMANDERING IS THE DELIBERATE DESIGN OF ELECTORAL DISTRICT BOUNDARIES TO FAVOR ONE POLITICAL PARTY.

SAME VOTERS → **DIFFERENT BOUNDARIES** → **DIFFERENT RESULTS**

OUR QUESTION:

HOW MUCH ELECTORAL ADVANTAGE CAN BOUNDARY DESIGN CREATE WHILE SATISFYING THE REQUIRED CONSTRAINTS?

PROJECT GOALS

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PROCESS

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PROJECT RESULT

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CONCLUSION

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QUESTION & ANSWER

THANK YOU